

DATA PAPER

Grassland aboveground biomass, composition, and chemistry from multiple sites and years in Yellowstone National Park

Douglas A. Frank¹ | Rick L. Wallen²

¹Department of Biology, Syracuse University, Syracuse, New York, USA

²Yellowstone Center for Resources, National Park Service, Yellowstone National Park, Wyoming, USA

Correspondence

Douglas A. Frank
Email: dafrank@sy.edu

Funding information

Division of Environmental Biology,
Grant/Award Number: 2143437

Handling Editor: Kathryn L. Cottingham

Abstract

This data set includes measurements of aboveground plant biomass (in grams per square meter), percent alive and dead, composition (percent graminoid [grasses, sedges, rushes] and non-graminoid [other monocots, dicots]), and carbon and nitrogen content (in parts per million) of aboveground biomass collected during three studies (1988 and 1989; 1999–2001; 2012–2014) at grasslands grazed by herds of elk (*Cervus canadensis*), bison (*Bison bison*), and pronghorn (*Antilocapra americana*) in Yellowstone National Park. A total of 25 different grasslands were sampled during the studies. At each grassland, measurements were made outside and inside small (1.5 × 1.5 m) temporary exclosures moved approximately monthly throughout each growing season to determine ungulate consumption and aboveground production. Plant data were also gathered at a subsample of 13 of the grasslands inside permanent exclosures erected during the summer before each study. Monthly aboveground plant P content (in parts per million) is also provided at six sites in 2013 and 2014. Location (latitude, longitude), elevation, and 0 to 10 cm total soil C and N are included for all the sites. There are no copyright or proprietary restrictions on the data; please cite this data paper when using the data in other works.

KEYWORDS

graminoid and non-graminoid composition, grazed grassland, plant aboveground biomass, plant aboveground nitrogen and phosphorus content, soil carbon and nitrogen content, ungrazed grassland, ungulate exclosures, Yellowstone National Park

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data are available in [Data S1](#) of the [Supporting Information](#) to this paper and are also available in Dryad at <https://doi.org/10.5061/dryad.6djh9w1f4>.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](#) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Ecology* published by Wiley Periodicals LLC on behalf of The Ecological Society of America.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Frank, Douglas A., and Rick L. Wallen. 2025. "Grassland Aboveground Biomass, Composition, and Chemistry from Multiple Sites and Years in Yellowstone National Park." *Ecology* 106(9): e70198. <https://doi.org/10.1002/ecy.70198>